HW # 0

Jan. 7, 2025

1) (1) To make peanut butter, blend 3/4 tsp. of table salt (NaCl) with 1 lbm of unsalted dry roasted peanuts. Assume 1 tsp. table salt weighs 6.0 g. The mixture is then ground into peanut butter. This recipe must be altered when unsalted peanuts are not available so that there is the same amount of salt in the final mixture. Lightly salted and salted peanuts are just unsalted peanuts with some salt already added. The sodium content of dry roasted peanuts are given below.
Unsalted: 5 mg Na/oz. (Notice: Unsalted peanuts contain some sodium.)
Lightly salted: 75 mg Na/oz.
Salted: 150 mg Na/oz.
Please notice that the data above give only the mass of sodium per mass of peanut/salt mix.
(a) How much table salt in tsp. needs to be added to 1 lbm of lightly salted peanuts to give the same salt content as in the original recipe for the final mixture?
(b) How much table salt in tsp. needs to be added to 1 lbm of salted peanuts to give the same salt content as in the original recipe for the final mixture?
(c) Comment on your results.

=16 oz

a) $\frac{0.75 \cancel{\text{tsp}} \text{ NaCl}}{1 \cancel{\text{lbm}} \text{ unsalted peanuts}} \times \frac{1 \cancel{\text{lbm}}}{16 \text{ oz}} \times \frac{6.0 \text{ g}}{1 \cancel{\text{tsp}}} = 0.28125 \text{ NaCl g/oz peanuts}$

$\frac{0.28125 \cancel{\text{NaCl g}}}{\text{oz peanuts}} \times \frac{1 \cancel{\text{mol NaCl}}}{58.44 \cancel{\text{g NaCl}}} \times \frac{\cancel{1 \text{mol Na}}}{1 \cancel{\text{mol NaCl}}} \times \frac{22.99 \text{ g Na}}{1 \cancel{\text{mol Na}}} = 0.11064 \frac{\text{g Na}}{\text{oz}}$

$\frac{5 \cancel{\text{mg}} \text{ Na}}{\text{oz peanuts}} \times \frac{1 \text{ g}}{10^3 \cancel{\text{mg}}} = 0.005 \frac{\text{g Na}}{\text{oz peanuts}}$

Total Na in salted butter w/ unsalted peanuts:

$0.11064 \frac{\text{g Na}}{\text{oz}} + 0.005 \frac{\text{g Na}}{\text{oz}} = 0.11564 \frac{\text{g Na}}{\text{oz}}$

Lightly Salted:

$\frac{75 \text{ mg Na}}{\text{oz peanuts}} \times \frac{1 \text{ g}}{10^3 \text{ mg}} = 0.075 \frac{\text{g Na}}{\text{oz}}$

$0.11564 \frac{\text{g Na}}{\text{oz}} - 0.075 \frac{\text{g Na}}{\text{oz}} = 0.0406 \frac{\text{g Na}}{\text{oz}}$ needs to be added

$0.0406 \frac{\text{g Na}}{\text{oz}} \times \frac{1 \text{ mol Na}}{22.99 \text{ g Na}} \times \frac{1 \text{ mol NaCl}}{1 \text{ mol Na}} \times \frac{58.44 \text{ g NaCl}}{1 \text{ mol NaCl}} = 0.1033 \frac{\text{g NaCl}}{\text{oz peanuts}}$

$\frac{0.1033 \text{ g NaCl}}{\text{oz peanuts}} \times \frac{1 \text{ tsp}}{6.0 \text{ g}} \times \frac{16 \text{ oz}}{1 \text{ lbm}} = 0.275 \frac{\text{tsp NaCl}}{1 \text{ lbm peanuts}}$

b) Salted:

$$\frac{150\text{ mg Na}}{\text{Oz peanuts}} \times \frac{1\text{g}}{10^3\text{mg}} = 0.150\frac{\text{g Na}}{\text{Oz}}$$

$$0.11564\frac{\text{g Na}}{\text{Oz}} - 0.150\frac{\text{g Na}}{\text{Oz}} = -0.03436\frac{\text{g Na}}{\text{Oz}}$$

You Can't make this recipe there's so much Na Can't be made w/ salt and peanuts

C) For the lightly salted you would need to add some amount of salt $\approx 0.28 \frac{\text{tsp NaCl}}{\text{Oz Peanuts}}$ and for the fully salted the recipe can not be made because there's so much Na in the Peanuts then needed.

R. 1: 20/20

2) (2) One hundred grams of solid copper is reacted with chlorine gas to form cupric chloride ($CuCl_2$) and cuprous chloride ($CuCl$). All the copper reacts. The combined mass of the cupric chloride and cuprous chloride formed is 200 g. What is the mass ratio of the cupric chloride to the cuprous chloride in the product?

$Cu_{(s)} + Cl_{(g)} \longrightarrow CuCl$ ✓

$Cu_{(s)} + Cl_{(g)} \longrightarrow CuCl_2$ ✓

Molar masses:

Cu: 63.54g/mol

Cl: 35.45g/mol

CuCl: 98.99g/mol

$CuCl_2$: 134.44g/mol

Mass ratio of Cu: to Product:

$CuCl = \frac{63.54}{98.99} = 0.6419$ ✓

$CuCl_2 = \frac{63.54}{134.44} = 0.4726$ ✓

Equations:

① $0.6419\, m_{CuCl} + 0.4726\, m_{CuCl_2} = 100g\ Cu$ ✓

② $m_{CuCl} + m_{CuCl_2} = 200g$ ✓

2 unknowns
2 equation

$m_{CuCl} = 200g - m_{CuCl_2}$

$0.6419(200g - m_{CuCl_2}) + 0.4726\, m_{CuCl_2} = 100g\ Cu$

$128.377g - 0.6419\, m_{CuCl_2} + 0.4726\, m_{CuCl_2} = 100g\ Cu$

$-0.1693\, m_{CuCl_2} = -28.377\ g\ Cu$ ✓

$m_{CuCl_2} = 167.66g$

$167.66 + m_{CuCl} = 200g$ $\quad m_{CuCl} = 32.34g$ ✓

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Mass Ratio: $\frac{CuCl_2}{CuCl} = \frac{167.66g}{32.34g} = 5.18$ ← Final Answer ✓

3)

(3) The ideal gas isobaric heat capacity $C_P(T)$ is a function of temperature only. Experimental data can be fit to function of temperature to give expressions such as:

$$C_P(T) = A + BT + CT^2 + DT^3$$

where the temperature is in Kelvin giving the heat capacity in J/mol-K. For oxygen gas at low pressure, this equation has the following parameter values: A = 28.11, B = -3.68e-6, C = 1.746e-5, and D = -1.065e-8 according to Table B.17 on page 665 of the textbook by Murphy. Check these values in your copy of the textbook. Some printings have the wrong values.

(a) Find the enthalpy change from 300K to 700K for oxygen using the $C_P(T)$ expression above. Give your result in J/mol.

(b) Make a change of the variable of integration such that $t = T/1000$. Since T is in units of Kelvin, then t has units of "kiloKelvin." Convert the heat capacity $C_P(T)$ to $C_P(t)$. Then do the appropriate integral to get the enthalpy change from 300K to 700K for oxygen. Again, give your result in J/mol.

Recall: $\Delta H = \int_{T_1}^{T_2} C_p \, dT$ ✓

a) $\Delta H = \int_{300K}^{700K} (28.11 - 3.68\times10^{-6}\,T + 1.746\times10^{-5}\,T^2 - 1.065\times10^{-8}\,T^3)\,dT$ ✓

$$= \left[28.11\,T - \frac{3.68\times10^{-6}}{2}T^2 + \frac{1.746\times10^{-5}}{3}T^3 - \frac{1.065\times10^{-8}}{4}T^4\right]\Bigg|_{300}^{700}$$ ✓

$$= 28.11(700-300) - \frac{3.68\times10^{-6}}{2}(700^2 - 300^2) + \frac{1.746\times10^{-5}}{3}(700^3 - 300^3) - \frac{1.065\times10^{-8}}{4}(700^4 - 300^4)$$ ✓

$= 12464.7$ J/mol ✓

B →

b) $t = \frac{T}{1000}$ $t_i = \frac{300}{1000} = 0.3$ kiloKelvin

$t_f = \frac{700}{1000} = 0.7$ kilokelvin ✓

Cp units : $\frac{Cp\,J}{mol \cdot K} \times \frac{1000\,K}{1\text{ kiloKelvin}} = 1000\,Cp = J/mol\text{ kiloK}$

$$\int_{0.3}^{0.7} 1000\,Cp\,dT$$

$*\, T = t(1000)$

$$1000\int_{0.3}^{0.7}\left(28.11 - 3.68\times10^{-6}t(1000) + 1.746\times10^{-5}t^2(1000)^2 - 1.065\times10^{-8}t^3(1000)^3\right)dT$$ ✓

$$1000\left[28.11\,t - 3.68\times10^{-6}\frac{t^2}{2}(1000) + 1.746\times10^{-5}\frac{t^3}{3}(1000)^2 - 1.065\times10^{-8}\frac{t^4}{4}(1000)^3\right]\Bigg|_{0.3}^{0.7}$$ ✓ ✓

$= \boxed{12464.7\ J/mol}$ ✓

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4)

(4) Wolfram Alpha gives the following result for a definite integral:

$$\int_1^h \frac{1}{40-4\sqrt{x-1}}\,dx = -\frac{\sqrt{h-1}}{2} - 5\log\left(10-\sqrt{h-1}\right) + 5\log(10)$$

for $\mathrm{Re}(h) \leq 101 \vee h \notin \mathbb{R}$

Work the integral yourself. Carefully show all the steps and needed changes of variable. Simplify the integral so that it only contains simple integrals such as:

$$\int \frac{1}{x}\,dx = \log(x) + \text{constant} \qquad \int x^n\,dx = \frac{x^{n+1}}{n+1} + \text{constant}$$

where log is the natural logarithm and n is any power.

$$\int_1^h \frac{1}{40-4\sqrt{x-1}}\,dx$$

$(u)^2 = (\sqrt{x-1})^2$

$u^2 = x-1$

$x = u^2+1$

$x = u^2+1$

$dx = 2u\,du$

when $x=1$ $u=0$

$x=h$ $u=\sqrt{h-1}$

$$\int_0^{\sqrt{h-1}} \frac{2u\,du}{40-4u} = \frac{1}{2}\int_0^{\sqrt{h-1}} \frac{u\,du}{10-u} = \frac{1}{2}\int_0^{\sqrt{h-1}} \frac{u+10-10}{10-u}\,du = \frac{1}{2}\int_0^{\sqrt{h-1}} \frac{(u-10)+10}{10-u}\,du$$

$$= \frac{1}{2}\int_0^{\sqrt{h-1}} \frac{u-10}{10-u} + \frac{10}{10-u}\,du = \frac{1}{2}\int_0^{\sqrt{h-1}} \underbrace{\cancel{\frac{-(10-u)}{10-u}}}_{=-1} + \frac{10}{10-u}\,du$$

$$= \frac{1}{2}\int_0^{\sqrt{h-1}} -1 + \frac{10}{10-u}\,du = \frac{1}{2}\left[-u - 10\ln|10-u|\right]\Big|_0^{\sqrt{h-1}}$$

$$= \frac{1}{2}\left[-\sqrt{h-1} - 10\ln|10-\sqrt{h-1}| - \left((-0) - 10\ln|10-0|\right)\right]$$

$$= \frac{1}{2}\left[-\sqrt{h-1} - 10\ln|10-\sqrt{h-1}| + 10\ln|10|\right]$$

$$= \frac{-\sqrt{h-1}}{2} - 5\ln|10-\sqrt{h-1}| + 5\ln|10|$$

Pr. 4: 20/20

5) (5) If you ask Google "How many calories are in ethanol?" Google will answer 7 per gram. These are dietary calories which are kilocalories. Hence, 7 kcal/g. The enthalpy of combustion for ethanol is about -1,235 kJ/mol for ethanol in the gas phase and -1,367 kJ/mol in the liquid state. These values are for 25℃ and 1 atm. Convert the 7 kcal/g to kJ/mol and compare with the two standard enthalpies. What conclusions can you make? Explain the minus signs on the standard ethalpies.

$$\frac{7\ k\cancel{cal}}{\cancel{g}} \times \frac{4.184\ kj}{1\ k\cancel{cal}} \times \frac{46.07\cancel{g}\ ethanol}{1\ mol} = 1{,}349.30\ kj/mol$$

ethanol gas $\Delta H = -1235$ kj/mol
ethanol liquid $\Delta H = -1367$ kj/mol
dietary $\Delta H = 1349$ kj/mol

Pr. 5: 20/20

The gas and liquid phases are negative which means heat is coming out or being released while the dietary one is positive meaning heat is coming in or being "consumed" by the system.